GIT Branch , Merge & Merge Conflict

What is Git Branching?

In Git, a `branch` is a new/separate version of the main repository.

Branches allow you to work on different parts of a project without impacting the main branch. When the work is complete, a branch can be merged with the main project.

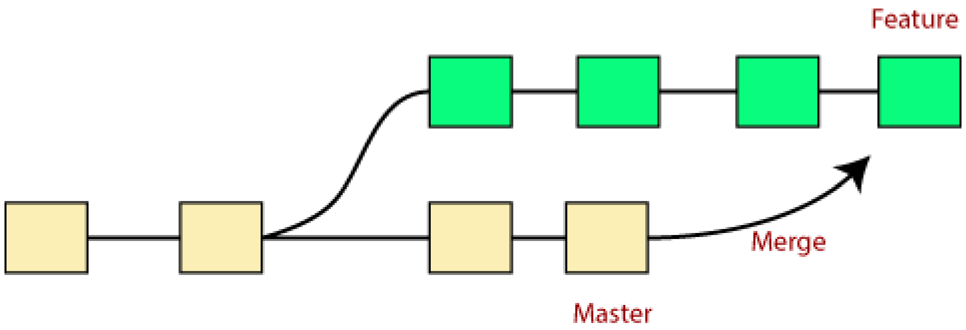
You can even switch between branches and work on different projects without them interfering with each other.

Branching in Git is very lightweight and fast!

Git Merge

In Git, the merging is a procedure to connect the forked history. It joins two or more Development history together. The git merge command facilitates you to take the data

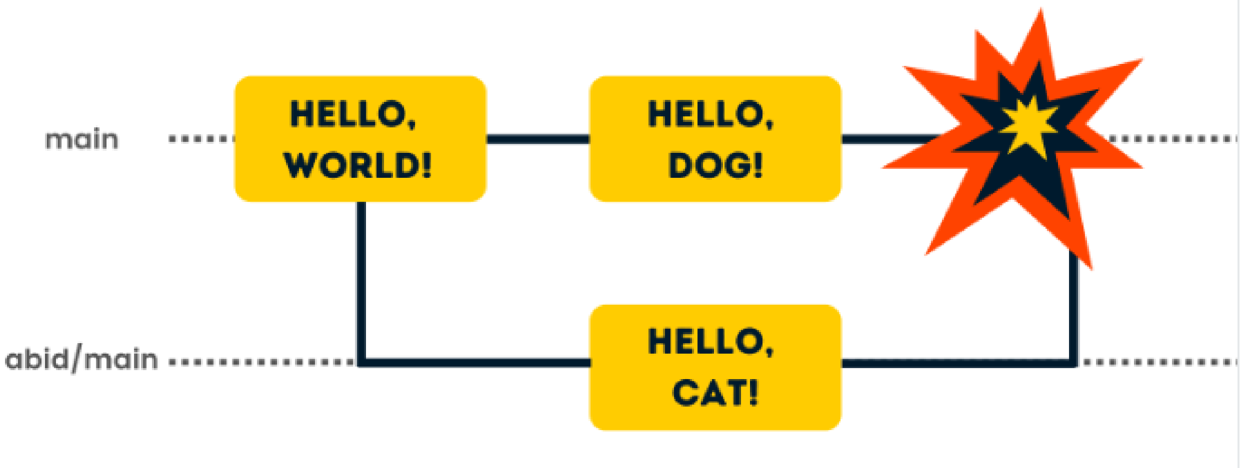
Created by git branch and integrate them into a single branch. Git merge will associate a series of commits into one unified history. Generally, git merge is used to combine two branches.



Merge Conflict

A merge conflict is an event that takes place when Git is unable to automatically resolve differences in code between two commits. Git can merge the changes automatically only if the commits are on different lines or branches.

In every situation where work can be parallelized, work will eventually overlap. Sometimes two developers will change the same line of code in two different ways; in such a case, Git can't tell which version is correct—that's something only a developer can decide.



Conflicts generally arise when two people have changed the same lines in a file, or if one developer deleted a file while another developer was modifying it. In these cases, Git cannot automatically determine what is correct.

Git fetch

The git fetch command retrieves commits, files, branches, and tags from a remote repository. Git isolates the fetched content from the local code. Therefore, the fetch provides a safe way to review the information before committing to your local branch.